

TECHNICAL PRINCIPLES	7.2 How can materials and processes be used to make final prototypes?	
	a. Understand how to select and safely use common workshop tools, equipment and machinery to manipulate materials by methods of: i. wasting/subtraction processes such as cutting, drilling, turning, milling ii. addition processes such as soldering, brazing, welding, adhesives, fasteners iii. deforming and reforming processes such as bending, vacuum forming.	
	b. Demonstrate an understanding of the role of computer-aided manufacture (CAM) and computer-aided engineering (CAE) to fabricate parts of a final prototype: i. additive manufacturing (3D printing) to fabricate a usable part ii. subtractive CNC manufacturing such as laser/plasma cutting, milling, turning and routing.	
	c. Demonstrate an understanding of measuring instruments and techniques used to ensure that products are manufactured accurately or within tolerances as appropriate.	
	d. Understand how the available forms, costs and working properties of materials contribute to the decisions about suitability of materials when developing and manufacturing their own products.	